

Measuring Thematic Drift in Computational Linguistics:

Alignment of TACL Articles with Journal Aims & Scope (2015–2025)

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Abstract

In this project, I looked at how research in computational linguistics and NLP has changed over time, and whether articles published in *Transactions of the Association for Computational Linguistics* (TACL) are still aligned with the journal's stated Aims & Scope. I collected 620 TACL articles from 2015 to 2025 and used a domain-specific Transformer model (SPECTER) to turn each paper into a vector. I then measured how close each paper is to an embedded version of the journal's official Aims & Scope using cosine similarity. The results show that the journal is broadly on-scope (mean score = 0.793) and that alignment is mostly stable over time rather than showing strong drift. Clustering the papers revealed three broad research topics: Traditional NLP, Deep Learning NLP, and LLMs / Generative AI, with some shifts in their relative shares after 2022.

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1 Introduction

NLP has changed a lot in recent years. The field went from classical methods like rule-based parsing and bag-of-words models, to deep learning, and then to large language models. This raises an interesting question: do the articles published in major NLP journals still match the official mission of those journals?

In this project, I focus on **thematic drift** — the idea that a journal’s research output can gradually move away from its stated aims, either because new topics emerge or because older priorities shrink. Instead of just reading papers and forming an opinion, I wanted to measure this drift in a structured way using a journal-specific dataset and semantic embeddings.

My three main research questions are:

1. How well do articles published in *Transactions of the Association for Computational Linguistics* align with the journal’s stated Aims & Scope?
2. Has this alignment changed from 2015 to 2025?
3. What are the main research topics in the corpus, and how have their shares shifted over time?

2 Methodology

2.1 Data Collection

I used the OpenAlex API to collect articles published in *Transactions of the Association for Computational Linguistics* (TACL). The dataset covers the years 2015 to 2025 and contains 620 journal articles with usable abstracts. This source was chosen because TACL is a journal focused on computational linguistics and natural language processing, which makes it appropriate for a journal-level alignment analysis.

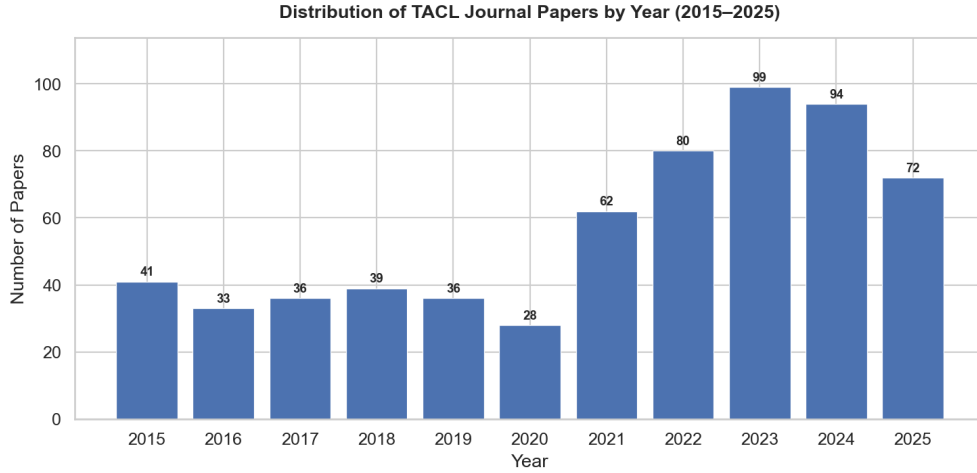


Figure 1. TACL articles per year in the dataset. The dataset contains 620 articles from 2015 to 2025, with larger yearly counts after 2021.

Each article was stored with its title, abstract, publication date, year, and journal category. Articles with very short abstracts (under 80 characters) or duplicate identifiers were removed. Since the number of available articles is not identical across years, I interpret the yearly trends carefully rather than treating each year as an equally sized sample.

2.2 Text Processing

I combined each article’s title and abstract into one text field, separated by a period. This matches the format that the SPECTER model was pre-trained on. I kept cleaning minimal — just unicode normalisation, lowercasing, and URL removal. I deliberately did not remove stop words, because transformer models rely on full sentence structure to understand meaning.

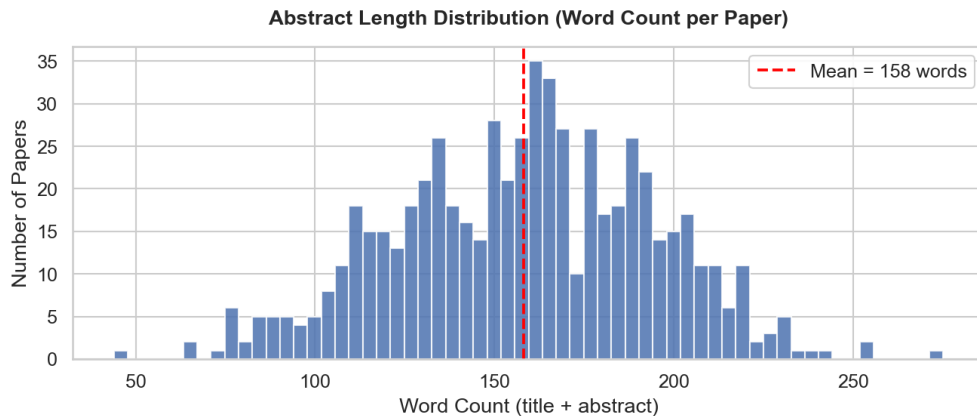


Figure 2. Word count distribution across articles (title + abstract). The mean is 158 words, which is suitable for the embedding model.

2.3 Embedding Model

I used **SPECTER** (`allenai-specter`), a BERT-based model trained specifically on scientific paper abstracts using citation graph context[6]. It produces 768-dimensional vectors. I chose it over a generic model like `all-MiniLM-L6-v2` because general models are trained on web text and may not handle scientific vocabulary as well. For example, the word “transformer” means something very specific in NLP, and a general model might not capture that context as clearly.

Embeddings were computed on CPU with a batch size of 8 and saved to disk as a `.npy` file so I would not have to recompute them every run.

2.4 Alignment Scoring

To measure how on-scope each article is, I needed a reference point. I used the official **Aims & Scope** of TACL as the journal’s thematic benchmark. The scope statement describes TACL as a journal for work in all areas of computational linguistics and natural language processing. I embedded this text using the same SPECTER model as the article abstracts. This gives me a single 768-dimensional “target vector” that represents the journal’s intended thematic focus. In this project, the scope text is used as a benchmark rather than a perfect manually labelled ground truth.

I then computed the cosine similarity between each article and this target vector:

$$\text{align}(i) = \frac{\mathbf{p}_i \cdot \mathbf{a}}{\|\mathbf{p}_i\| \cdot \|\mathbf{a}\|} \quad (1)$$

where $\mathbf{p}_i$ is the paper embedding and $\mathbf{a}$ is the scope embedding. Higher values mean the article is more thematically aligned with the journal.

I then classified papers using z-scores:

Category	Criterion	Meaning
High alignment	$z > +1.0$	Strongly on-scope
Aligned	$-1.5 \leq z \leq +1.0$	Normal range
Outlier	$z < -1.5$	Significantly lower alignment

Table 1. Z-score thresholds for article classification.

2.5 Clustering

To find the main research topics, I applied KMeans clustering ($k=3$) directly on the 768-dimensional embeddings. I chose $k=3$ based on silhouette scores evaluated across $k \in \{2, 3, 4, 5\}$. I used UMAP only for the 2D visualisation — the clustering itself was done

in the full embedding space. I labelled the clusters manually by looking at representative articles from each group.

Topic trends were tracked using yearly market share:

$$\text{share}(k, y) = \frac{n_{k,y}}{\sum_{k'} n_{k',y}} \quad (2)$$

This normalises by year so that changes in publication volume do not hide changes in research focus.

3 Results

3.1 Alignment Distribution

The mean alignment score across all 620 articles is $\mu = 0.793$. Figure 3 shows the full distribution.

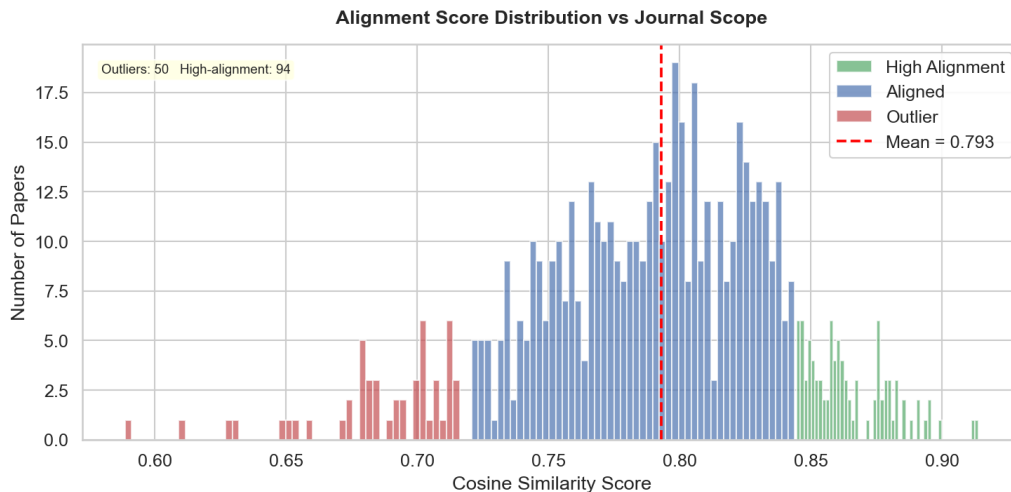


Figure 3. Alignment score distribution. 50 articles (8.1%) are outliers, 94 articles (15.2%) are high-alignment, and the remaining 76.8% fall in the normal aligned range. Mean = 0.793.

Most articles fall between 0.75 and 0.85, which suggests that ACL articles are generally close to the journal’s stated scope. The 50 outliers are not necessarily bad or irrelevant papers. They may be more interdisciplinary, use unusual terminology, or focus on methods and applications that are less explicitly reflected in the scope text. The 94 high-alignment articles are more directly connected to central NLP themes such as language modelling, translation, dialogue, parsing, and representation learning.

As a simple qualitative check, I inspected the articles that appeared near the top and bottom of the alignment ranking. The high-alignment examples mostly focus on central NLP themes, while the lower-alignment examples are less directly matched to the wording

of the scope. This supports the alignment score as a useful signal, but I do not treat it as a final judgement about whether a paper belongs in TACL.

Type	Article title	Year	Score	Inspection note	
High	Holmes: A Benchmark to Assess the Linguistic Competence of Language Models	2024	0.914	Central theme.	NLP
High	Efficient Methods for Natural Language Generation	2023	0.911	Central theme.	NLP
High	Quality at a Glance: An Audit of Web-Crawled Multilingual Datasets	2022	0.899	Central theme.	NLP
Low	Removing Backdoors in Pre-trained Models by Regularized Continual Pre-training	2023	0.589	Less direct scope match.	
Low	Learning End-to-End Goal-Oriented Dialog with Multiple Answers	2019	0.609	Less direct scope match.	
Low	Revisiting Few-shot Relation Classification	2021	0.629	Less direct scope match.	

Table 2. Examples of high- and low-alignment TACL articles inspected manually.

3.2 Thematic Drift Over Time

Figure 4 shows how the mean alignment score changed year by year.

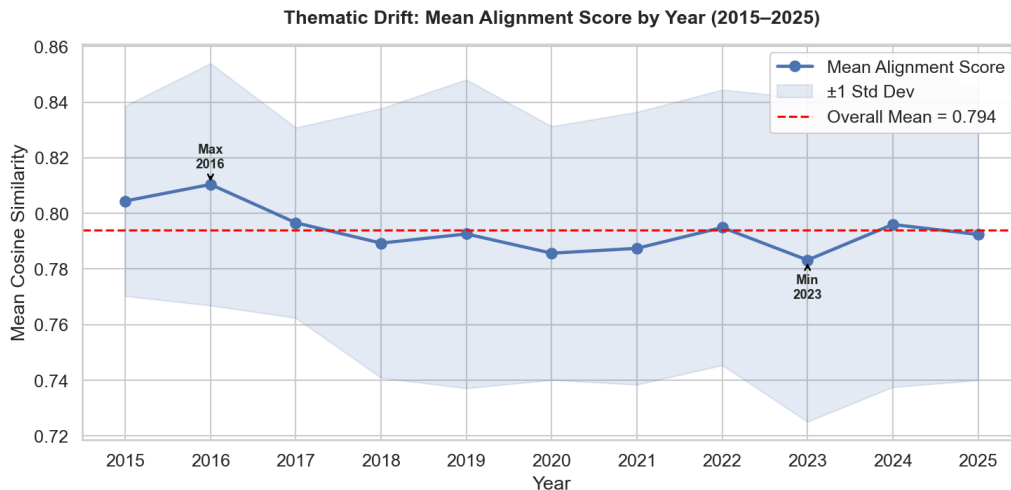


Figure 4. Mean alignment score per year with ± 1 standard deviation band. Alignment remains close to the overall mean from 2015 to 2025, with the highest yearly mean in 2016 and the lowest in 2023.

The trend is more stable than I expected. Given the rise of deep learning and LLMs, I expected to see a stronger shift away from the original journal scope. Instead, the mean alignment score stays close to the overall average across most years. This suggests that TACL has remained broadly aligned with its stated scope, even though the internal topics

of NLP have changed over time. The small drop around 2023 may reflect the rapid growth of newer LLM-related work, but the change is not large enough to claim strong thematic drift.

3.3 Research Topics

Figure 5 shows the UMAP projection of all articles coloured by their cluster.

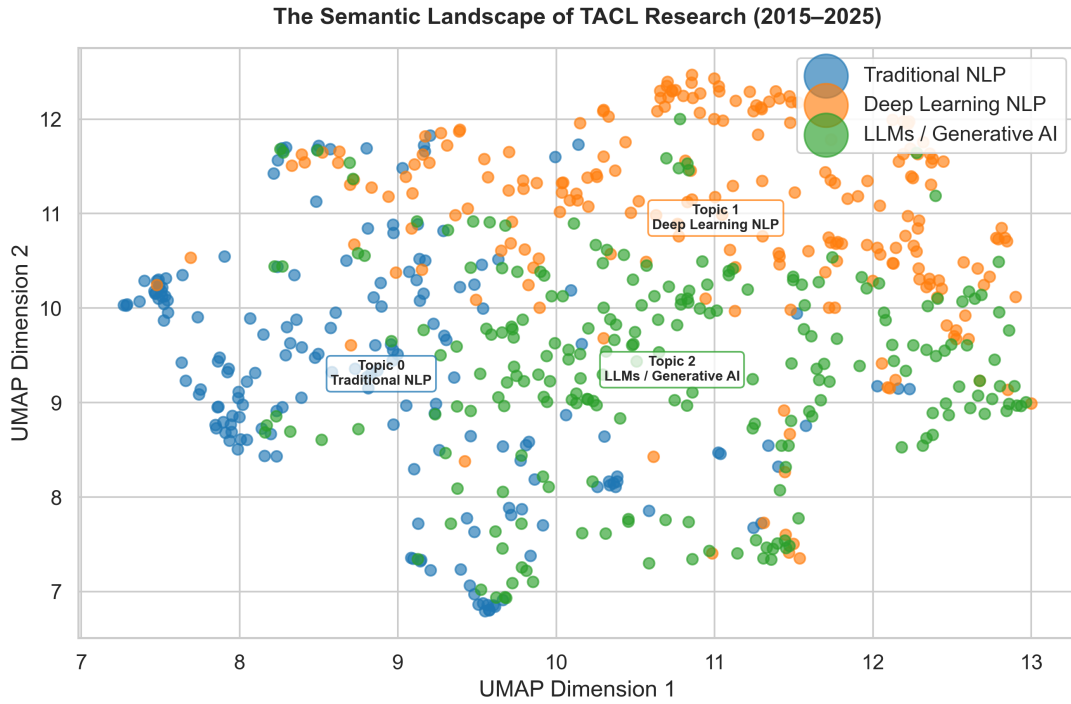


Figure 5. UMAP projection of article embeddings coloured by cluster. The three broad topics represent Traditional NLP, Deep Learning NLP, and LLMs / Generative AI.

The three clusters show overlapping but still meaningful regions. This makes sense because NLP topics are not completely separate: for example, a paper on machine translation can also involve deep learning, and a paper on prompting can still evaluate traditional NLP tasks. I labelled the clusters by manually checking representative titles and abstracts from each group.

Figure 6 shows how the share of each topic has changed over time.

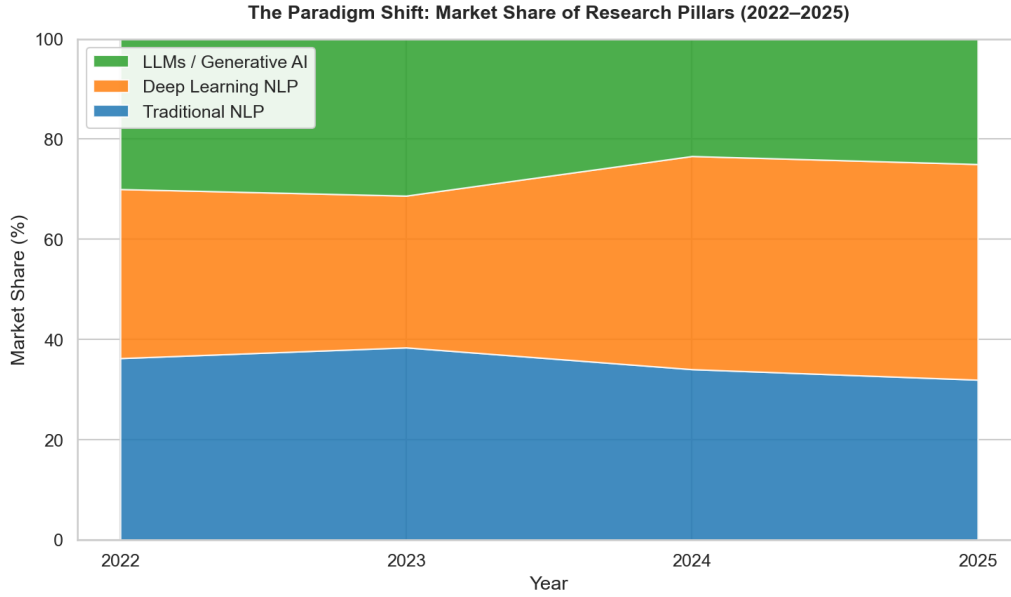


Figure 6. Yearly topic share of each research cluster from 2022 to 2025. Deep Learning NLP increases in share ($\sim 34\% \rightarrow \sim 43\%$), Traditional NLP decreases slightly ($\sim 36\% \rightarrow \sim 32\%$), and LLMs / Generative AI decrease modestly ($\sim 30\% \rightarrow \sim 25\%$).

Cluster	Share in 2022	Share in 2025
Traditional NLP	$\sim 36\%$	$\sim 32\%$
Deep Learning NLP	$\sim 34\%$	$\sim 43\%$
LLMs / Generative AI	$\sim 30\%$	$\sim 25\%$

Table 3. Approximate topic share in 2022 and 2025.

The most visible change is the growth of the Deep Learning NLP cluster from 2022 to 2025. Traditional NLP decreases slightly, but it does not disappear. This is important because many newer papers still evaluate classic tasks such as parsing, translation, summarisation, and coreference resolution. The LLMs / Generative AI cluster is also substantial, but it does not fully dominate the journal. Overall, the results suggest thematic evolution inside NLP rather than a complete replacement of older research directions.

4 Limitations and Future Work

There are a few important limitations to keep in mind.

The first limitation is the scope reference. I used the journal’s Aims & Scope as a single embedded text, but journal scope statements are usually short and general. A slightly different formulation of the same scope could produce a slightly different target vector. For this reason, the alignment scores should be interpreted as approximate semantic similarity scores rather than exact editorial judgements.

I also only looked at titles and abstracts, not full paper text. This is common in large-scale studies, but it does mean that some articles might be misrepresented if their abstract is very general or does not clearly describe the contribution.

The cluster labels (Traditional NLP, Deep Learning NLP, LLMs / Generative AI) were assigned by me manually. A more principled approach would extract TF-IDF keywords automatically from each cluster. The choice of $k=3$ is also a simplification — a larger k might reveal finer-grained sub-topics such as machine translation, dialogue systems, information extraction, or multilingual NLP.

Finally, I did not run any formal statistical tests on the drift trend. The yearly means are fairly stable, and I cannot rule out that small differences between years are within normal variation. In future work, I would add a regression test, try BERTopic for more interpretable clustering, and compare TACL with another NLP journal to see whether the same pattern appears.

5 Conclusion

This project built an automated pipeline to measure thematic drift in a real computational linguistics journal. Using SPECTER embeddings and cosine similarity, I compared 620 TACL articles from 2015 to 2025 to a journal scope reference vector. The main findings are that TACL is broadly on-scope (mean alignment = 0.793), alignment is mostly stable over time, and the biggest internal change is a shift in the relative share of Traditional NLP, Deep Learning NLP, and LLMs / Generative AI. The results do not show strong evidence that recent NLP developments have pushed TACL outside its stated scope. Instead, they suggest that the journal’s scope is broad enough to include both older NLP tasks and newer modelling paradigms.

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